

Stoke's Theorem

Statement of Stoke's theorem

If S is an open surface bounded by a simple closed curve C if $\vec{F}$ is continuous having continuous partial derivatives in S and C , then

$$\int_c \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

(or)

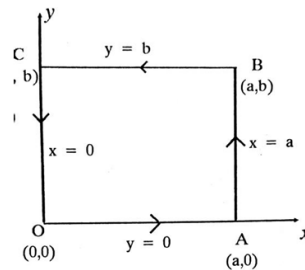
$$\int_c \vec{F} \cdot d\vec{r} = \iint_S \nabla \times \vec{F} \cdot \hat{n} \, ds$$

$\hat{n}$ is the outward unit normal vector and C is traversed in the anti-clockwise direction.

Problems

Example 1. Verify Stokes' theorem for a vector field defined by $\vec{F} = (x^2 - y^2)\vec{i} + 2xy\vec{j}$ in a rectangular region in the xy plane bounded by the lines $x = 0, x = a, y = 0, y = b$

Solution:



By Stokes theorem, $\int_c \vec{F} \cdot d\vec{r} = \iint_S \text{Curl } \vec{F} \cdot \hat{n} \, dS$

To evaluate: $\iint_S \text{Curl } \vec{F} \cdot \hat{n} \, dS$

Given $\vec{F} = (x^2 - y^2)\vec{i} + 2xy\vec{j}$

$\text{Curl } \vec{F} = \nabla \times \vec{F}$

$$\begin{aligned} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - y^2 & 2xy & 0 \end{vmatrix} \\ &= \vec{i}(0) - \vec{j}(0 - 0) + \vec{k}[2y - (0 - 2y)] \\ &= 4y \vec{k} \end{aligned}$$

Since the surface is a rectangle in the xy plane, $\hat{n} = \vec{k}$, $dS = dxdy$

$$\text{Curl } \vec{F} \cdot \hat{n} = 4y \quad \vec{k} \cdot \vec{k} = 4y$$

Order of integration is $dxdy$

x varies from $x = 0$ to $x = a$

y varies from $y = 0$ to $y = b$

$$\begin{aligned} \Rightarrow \iint_S \text{Curl } \vec{F} \cdot \hat{n} dS &= \int_0^b \int_0^a 4y dx dy \\ &= \int_0^b 4y [x]_0^a dy \\ &= \int_0^b 4ay dy \\ &= \left[\frac{4ay^2}{2} \right]_0^b \\ &= 2ab^2 \end{aligned}$$

$$\Rightarrow \iint_S \text{Curl } \vec{F} \cdot \hat{n} dS = 2ab^2 \quad \dots (1)$$

Here the line integral over the simple closed curve C bounding the surface $OABCO$ consisting of the edges OA , AB , BC and CO .

Curve	Equation	Limit
OA	$y = 0$	$x = 0$ to $x = a$
AB	$x = a$	$y = 0$ to $y = b$
BC	$y = b$	$x = a$ to $x = 0$
CO	$x = 0$	$y = b$ to $y = 0$

$$\text{Therefore, } \int_C \vec{F} \cdot d\vec{r} = \int_{OABCO} \vec{F} \cdot d\vec{r}$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_{OA} + \int_{AB} + \int_{BC} + \int_{CO}$$

$$\vec{F} \cdot d\vec{r} = (x^2 - y^2) + 2xydy \quad \dots (2)$$

On OA : $y = 0, dy = 0, x$ varies from 0 to a

$$(2) \Rightarrow \vec{F} \cdot d\vec{r} = x^2 dx$$

$$\begin{aligned} \int_{OA} \vec{F} \cdot d\vec{r} &= \int_0^a x^2 dx \\ &= \left[\frac{x^3}{3} \right]_0^a = \frac{a^3}{3} \end{aligned}$$

On AB : $x = a, dx = 0, y$ varies from 0 to b

$$(2) \Rightarrow \vec{F} \cdot d\vec{r} = 2ay \, dy$$

$$\begin{aligned} \int_{AB} \vec{F} \cdot d\vec{r} &= \int_0^b 2ay \, dy \\ &= \left[\frac{2ay^2}{2} \right]_0^b = ab^2 \end{aligned}$$

On BC : $y = b, dy = 0, x$ varies from a to 0

$$(2) \Rightarrow \vec{F} \cdot d\vec{r} = (x^2 - b^2) dx$$

$$\begin{aligned} \int_{BC} \vec{F} \cdot d\vec{r} &= \int_a^0 (x^2 - b^2) dx \\ &= \left[\frac{x^3}{3} - b^2 x \right]_a^0 \\ &= -\frac{a^3}{3} + ab^2 \end{aligned}$$

On CO : $x = 0, dx = 0, y$ varies from b to 0

$$(2) \Rightarrow \vec{F} \cdot d\vec{r} = 0$$

$$\int_{CO} \vec{F} \cdot d\vec{r} = 0$$

$$(2) \Rightarrow \vec{F} \cdot d\vec{r} = \frac{a^3}{3} + ab^2 - \frac{a^3}{3} + ab^2 = 2ab^2 \quad \dots (3)$$

$$\text{From (3) and (1)} \quad \int_c \vec{F} \cdot d\vec{r} = \iint_S \text{Curl } \vec{F} \cdot \hat{n} \, dS$$

Hence Stokes theorem is verified.

Example 2. Evaluate $\int \vec{F} \cdot d\vec{r}$ by Stokes' theorem, where $\vec{F} = y^2 \vec{i} + x^2 \vec{j} + (x + z) \vec{k}$, and C is the boundary of the triangle with vertices at $(0,0,0), (1,0,0)$ and $(1,1,0)$.

Solution:

$$\text{Stoke's theorem is } \int_c \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds \quad \dots (1)$$

$$\text{Given } \vec{F} = y^2 \vec{i} + x^2 \vec{j} + (x + z) \vec{k}$$

And C is triangle $(0, 0, 0), (1, 0, 0)$ and $(1, 1, 0)$.

Since z -coordinate of each vertex is zero the triangle lies in xy - plane with corners $(0, 0), (1, 0)$ and $(1, 1)$.

$$\text{To evaluate : } \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

In xy - plane $\hat{n} = \vec{k}$, $ds = dxdy$

$$\begin{aligned}\text{curl } \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & x^2 & -(x+z) \end{vmatrix} \\ &= \vec{i}(0) - \vec{j}(-1) + \vec{k}(2x-2y) \\ &= \vec{j} + 2(x-y)\vec{k} \\ \text{curl } \vec{F} \cdot \hat{n} &= (\vec{j} + 2(x-y)\vec{k}) \cdot \vec{k} \\ &= 2(x-y)\end{aligned}$$

Limits:

x varies from y to 1 .

y varies from 0 to 1 .

$$\begin{aligned}\therefore \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds &= \int_0^1 \int_y^1 2(x-y) dxdy \\ &= 2 \int_0^1 \left[\frac{x^2}{2} - xy \right]_y^1 dy \\ &= 2 \int_0^1 \left(\frac{1}{2} - y - \frac{y^2}{2} + y^2 \right) dy \\ &= 2 \left[\frac{y}{2} - \frac{y^2}{2} - \frac{y^3}{6} + \frac{y^3}{3} \right]_0^1 \\ &= 2 \left[\frac{1}{2} - \frac{1}{2} - \frac{1}{6} + \frac{1}{3} \right] \\ &= 2 \left[\frac{1}{6} \right] = \frac{1}{3}\end{aligned}$$

From (1), $\int_c \vec{F} \cdot d\vec{r} = \frac{1}{3}$

Exercise:

Q1. Evaluate $\int \vec{F} \cdot d\vec{r}$ by Stokes' theorem, where $\vec{F} = yz \vec{i} + xz \vec{j} + xy \vec{k}$. Ans: 0

Q2. Verify Stokes' theorem for a vector field defined by $\vec{F} = x^2 \vec{i} + xy \vec{j}$, integrated around the square in the $z = 0$ plane whose sides are along the lines $x = 0, y = 0, x = a, y = a$. Ans: $a^3/2$